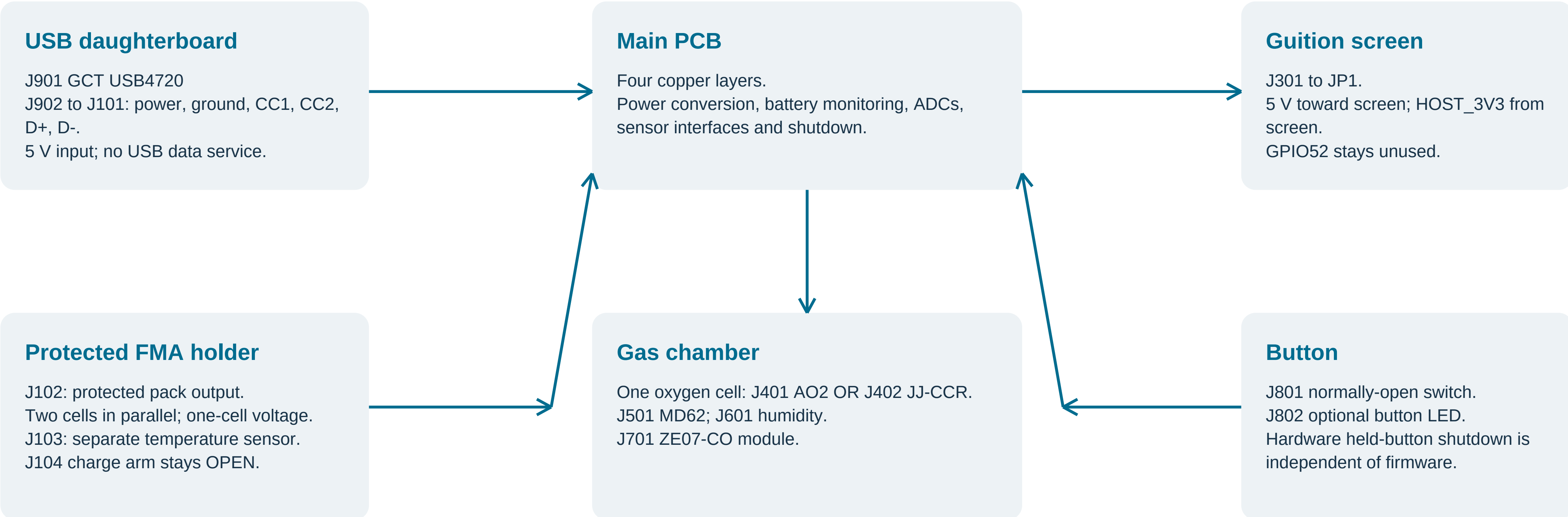
The complete connection map

Six logical harness groups connect the main PCB to the installed modules. Connector drawings are schematic views; no mating orientation is implied.



Before making a cable: identify both mating parts, locate pin 1 from the actual mating side, verify continuity without power, and record wire gauge, length, strain relief and minimum bend radius. The model does not yet establish every purchased connector.

Drawing sources: current KiCad net export and firmware/JP1 contract. Full measurements: mechanical/MEASUREMENTS.md. Electrical bring-up: electrical/measurement-and-bringup.md.

H01 - Guition JP1 to main J301

Logical pin-to-same-pin mapping. This is a numbered list, not a view of the two-row connector. Pitch, row spacing, mating height and cable exit still require measurement.

Main mate candidate: Samtec HTSW-113-07-L-D-007 / IDSD-13-S-04.00-G-P07. Configured availability pending.

Pin	Net	GPIO	Direction
1	HOST_3V3	-	from screen
2	HOST_5V	-	to screen
3	HOST_3V3	-	from screen
4	HOST_5V	-	to screen
5	GND	-	ground return
6	GND	-	ground return
7	NC - leave unused	-	no wire
8	POWER_KILL_N	33	host open drain
9	CO_UART_EN	51	host output
10	CO_UART_RX	31	host input
11	USB_ILIM_AUTH	50	host output
12	CO_UART_TX	30	host output
13	CHG_INT_N	49	host input

Pin	Net	GPIO	Direction
14	I2C_SCL	29	host output
15	NC - leave unused	-	no wire
16	GND	-	ground return
17	HE_ENABLE	34	host output
18	HOST_3V3	-	from screen
19	POWER_INT_N	32	host input
20	NC - leave unused	-	no wire
21	I2C_SDA	28	bidirectional
22	NC - leave unused	-	no wire
23	NC - leave unused	-	no wire
24	NC - leave unused	-	no wire
25	NC - leave unused	-	no wire
26	NC - leave unused	-	no wire

Power-entry HOLD: Guition ties JP1 5 V to the IP5306 boost output; external input at JP1 is still unqualified. Analyzer reverse isolation does not establish that use. Keep J301 disconnected for wired USB service; leave Guition BAT empty.
Supplies: 2/4 HOST_5V to screen; 1/3/18 HOST_3V3 from screen; 5/6/16 ground.
Pin 13 / GPIO49: shared charger IRQ and lost latch feedback. GPIO50 commands permission; GPIO49 checks it.

H02 - concealed bottom USB

J902 on the thin USB board connects to J101 on the main board. Pin numbering is logical: verify the silkscreen and continuity at both ends.

J902 pin	Net	J101 pin	Planned wire
1	USB_5V	1	24 AWG
2	GND	2	24 AWG
3	USB_CC1	3	28 AWG
4	USB_CC2	4	28 AWG
5	USB_D_P	5	28 AWG
6	USB_D_M	6	28 AWG

Electrical provisions

Two internal CC pull-downs in the sink controller; do not fit duplicate external Rd resistors.
D+/D- go only to the BC1.2 detector, not the BQ25895.
Upstream current limiting must work before firmware starts.
No fast-charge or USB-PD negotiation is provided.

Mechanical datums to verify

GCT drawing: 0.60 +/-0.10 mm board.
PCB nominal Z24.9-25.5; port centre Z26.0.
Nominal clearance above PCB to support bridge: 5.7 mm before solder/wire.
Check plug overmould, pigtail bend and strain relief physically.

Wire gauges are design proposals; qualify the actual insulation, solder process, length and rated current. Do not interpret a clear cable line here as a verified installed bend.

H03-H06 - sensors, pack and button

Only one oxygen cell is installed. J402 shield/shell carries the negative cell signal; do not bond it to chassis or ground.

Connector	Module	PCB pin-to-net map
J401	AO2	1: O2_A_RAW_P; 2: O2_A_RAW_N; 3: NC - leave unused
J402	JJ-CCR	1: O2_B_RAW_P; 2: O2_B_RAW_N
J501	MD62	1: HE_3V0; 2: HE_SENSE; 3: GND
J601	Humidity	1: BME_VIN; 2: GND; 3: BME_SCL; 4: BME_SDA
J701	ZE07-CO	1: CO_5V28; 2: GND; 3: CO_TX_CABLE; 4: CO_RX_CABLE
J102	Protected pack	1: PACK_P; 2: GND
J103	Pack NTC	1: PACK_TS; 2: GND
J801	NO button	1: Net-(J801-Pin_1); 2: GND
J802	Button LED	1: BUTTON_LED_A; 2: GND

The remote end remains a measurement gate. Check the actual module pin order, connector gender, cable colour/polarity and sensor lead drawing. A PCB net name alone does not establish the remote pin number. Humidity wiring must use a humidity-capable BME280; CO is not CO2.

J104 is a charge-arm link, not a sensor selector. Keep it open for commissioning. Oxygen choice and calibration are stored through the touchscreen; GPIO52/J301 pin 7 remains unused.

Measurements that unblock the order

Record millimetres, instrument accuracy, the actual part marking, and photographs from the mating side. Preserve the approved exterior while resolving these interfaces.

Item	Needed evidence
Guition JP1	First-to-last of 13 physical pins in one row; row spacing; pin 1; mated height.
Battery holder / cells	Actual plug and polarity; protrusions; cell maker/model and charging limits.
AO2 and JJ-CCR	Sealing shoulder to nose and rear; thread engagement; seal size and face.
J401 / J402 cables	Mated 90-degree elbow envelope, exit direction, grip and bend radius.
USB cartridge	Real connector on 0.60 mm coupon; notch/lands; wire bend and plug clearance.
Gas modules	MD62 leads; actual BME280 board; ZE07-CO connector and can height.
Chamber seals	Real gasket/feedthrough/fitting dimensions and supplier compression guidance.
Fasteners / button	Insert series/pilot; screw engagement; button nut and terminal stack.

Recorded already: display 69.3 x 116.8 x 13.7 mm; occupied protected holder 42 x 80.4 x 20.35 mm. The owner confirms a common oxygen thread, with JJ-CCR 2 mm smaller in diameter and 2 mm longer. The location of that extra length and the sealing faces still need verification.

Staged first power - after release gates close

All powered and physical checks are pending. Start with modules and cells disconnected, microscope inspection and unpowered continuity. Record test limits before applying power.

Stage	Check
1 Inspect	Check BOM/DNP, pin 1, solder bridges, exposed pads, rail resistance and every harness.
2 Inhibited rails	J104 OPEN. Current-limited supply; check charger inhibition and each rail before loads.
3 USB behavior	A-to-C / C-to-C, both orientations; cold attach, weak cable, detach and reattach.
4 Converters / host	Load steps, ripple, overshoot, reverse current, standby and Guition supply isolation.
5 Button / storage	Short/held press, stalled firmware, save interruption and safe battery disconnection.
6 Modules	Add one at a time. Check IDs, wiring, levels, ADC precision inputs and open/short faults.
7 Charging	Only with documented cells/NTC/protection: voltage, current, faults, termination and heat.
8 Reference gas	Condition MD62; characterize known mixtures, environment, drift and held-out validation.

Available: microscope, soldering iron, hot air, multimeter and current-limited bench supply. Obtain oscilloscope and controlled-load access for transients, ripple, inrush and bus timing. A supply that cannot sink current is not a battery emulator.

Retain board/firmware hashes, instrument identity/accuracy, hookup, limits, readings, conditions and corrective actions. O2 +/-0.2 percentage points and He +/-0.5 points remain unproven targets. Full procedure and source-linked limits are in electrical/measurement-and-bringup.md.